

BioStack Protocol Packet

Provider-review draft for phased metabolic, repair, and cognitive observability

Use boundary

This packet is an educational and observational planning artifact. It is not medical advice, not a prescription, and not a substitute for clinician oversight. All injectables, investigational compounds, lab interpretation, dose changes, and ultrasound findings should be reviewed with a qualified medical provider.

Quick-Reference Schedule

Weeks	Move	Protocol element	Decision rule
1-2	NAD+ only	25 mg SubQ, twice weekly	Establish baseline tolerance. Track energy AM/PM, sleep, mood, digestion, injection-site response.
3	Add MOTS-c	5 mg SubQ, 2x weekly	Only add if NAD+ is quiet. Keep one clear week before adding GHK-Cu.
4	Add GHK-Cu earlier	1 mg SubQ daily	Clean intervention week. Do not add BPC/KPV in the same week. Track skin, recovery, sleep, inflammation, mood, and site response.
5-6	Add repair layer	BPC-157 250 mcg daily. KPV 200-300 mcg daily.	Start low. Prefer adding BPC first, then KPV several days later if tolerance is excellent.
7	Stabilize	NAD+ 75-100 mg 2-3x weekly. Continue MOTS-c, GHK-Cu, repair layer.	No new major additions unless all prior layers are stable.
8-10	Retatrutide gate	Provider-approved low start only	Do not start unless labs are reviewed, GI status is stable, hydration is solid, and no unresolved adverse signals exist.
11-13+	Cognitive layer	Semax/Selank only after retatrutide is stable	Avoid adding cognitive peptides while appetite, sleep, mood, GI, or heart-rate signals are shifting.
14+	Optional Dihexa	Short cautious cycles only	Highest caution item. Do not add unless the rest of the stack is boringly stable and provider agrees.

Design principle

The goal is not to add everything fast. The goal is to preserve attribution. One new variable at a time gives BioStack clean data and gives your provider a real signal instead of a supplement crime scene.

Decision Gates

Gate	Continue only if
Add MOTS-c	NAD+ has no persistent adverse signal and sleep, digestion, mood, and injection-site response are acceptable.
Add GHK-Cu	NAD+ plus MOTS-c are stable. GHK-Cu gets its own observation window before BPC/KPV.
Add BPC/KPV	No unresolved inflammation, sleep, heart-rate, mood, GI, or injection-site issue.
Add retatrutide	Recent labs reviewed, provider approval documented, no active GI issue, hydration/electrolytes are stable, resting heart rate acceptable.
Add cognitive peptides	Retatrutide is stable and not causing sleep disruption, mood changes, anxiety, appetite over-restriction, or GI issues.

Phase 1: NAD+ Foundation and Mitochondrial Build, Weeks 1-6

Goal: restore the metabolic and redox baseline first, then layer mitochondrial support after NAD+ tolerance is clear. This phase should feel intentionally boring at first. Boring is data.

- Weeks 1-2: NAD+ only at the initial low dose. Do not add MOTS-c, repair peptides, or cognitive peptides yet.
- Week 3: Add MOTS-c only if NAD+ tolerance is clearly good.
- Weeks 5-6: Move toward the higher NAD+ maintenance range only if the lower ranges are well tolerated.
- Track: energy AM/PM, sleep quality, clarity/mood, digestion, injection-site response, training recovery, resting heart rate if available.

Phase 2: Early GHK-Cu and Repair Layer, Weeks 4-7

Goal: introduce GHK-Cu earlier as its own clean intervention, then add BPC-157 and KPV after attribution is preserved. This is the main revision from the original schedule.

- Week 4: Introduce GHK-Cu as a standalone layer after NAD+ and MOTS-c are stable.
- Week 5 or 6: Add BPC-157 and KPV after GHK-Cu has a quiet observation window.
- Rationale: GHK-Cu is a reasonable earlier add because it is less likely to dominate appetite, glucose, or GI signals than retatrutide. It still deserves its own observation window.
- Avoid: starting GHK-Cu, BPC-157, and KPV all in the same week. Mechanistically attractive, observability nightmare.

Phase 3: Metabolic Optimization with Retatrutide, Weeks 8-13+

Goal: introduce retatrutide only after the metabolic and repair base is stable. Retatrutide is the most signal-dominant compound in the packet and should not be treated like just another small add-on.

- Do not start before provider review, recent labs, and stable GI status.
- Track appetite, hydration, constipation/diarrhea, nausea, reflux, abdominal pain, sleep, mood, resting heart rate, training performance, and weight trend.
- If liver or lipid markers worsen, treat that as a review trigger rather than pushing through.

Phase 4: Cognitive Enhancement, Weeks 11-15+

Goal: add nootropic or neuroprotective peptides only after the metabolic layer is not actively moving the goalposts.

- Semax/Selank: consider only once retatrutide is stable and sleep/mood are clean.
- Dihexa: highest caution. Keep later, short, and provider-supervised. Do not introduce while other variables are moving.

Support Stack and Meal Anchors

Important wording change		
The support stack below is listed as a provider-review support plan. Some mechanisms are plausible and useful, but not all claims are proven for this exact protocol. The strongest version is to track objective labs and symptoms instead of assuming protection.		
Support item	Planned amount	Purpose and caution
First meal	30-40 g protein + about 40 g clean carbs	Supports muscle protein synthesis, hepatic export capacity, and better tolerance during appetite suppression.
L-carnitine	3 g daily with food	Fatty-acid transport support. If triglyceride/HDL ratio worsens, discuss whether increasing toward 4 g/day is appropriate.
Phosphatidylcholine	3 g daily with food	Cell membrane and bile/lipid handling support. Use especially cautiously if combining multiple lipid-active interventions.
Vitamin E	400 IU daily	Antioxidant support for lipid membranes. Caution: NIH notes bleeding interactions and mixed long-term trial signals at higher supplemental doses. Provider review is important if using anticoagulants, antiplatelets, surgery, or prostate cancer risk concerns.
Selenium	200 mcg daily	Antioxidant enzyme cofactor support. Keep total intake below the adult upper limit of 400 mcg/day unless provider-directed.
TUDCA	500 mg with each of the 2 biggest meals, total 1000 mg/day	Bile-acid support candidate during retatrutide. Strongly word as support, not guaranteed prevention. Review with provider, especially with

		gallbladder, bile duct, liver disease, or medication issues.
NAC	1000 mg daily	Glutathione precursor support and oxidative-stress support. Review if taking nitroglycerin, anticoagulants, or if asthma/GI sensitivity is present.
Methyl B complex, B12, folate	Per label/labs/provider	Methylation and liver-support cofactors. Best guided by B12, folate, homocysteine if available, CBC, and symptoms.
Magnesium glycinate	500 mg nightly	Sleep, muscle, glucose, and enzymatic cofactor support. Adjust if diarrhea, kidney disease, or sedating meds are present.
Zinc	30 mg daily with food	Immune and hepatic enzyme cofactor support, including alcohol dehydrogenase biology. Avoid long-term high-dose zinc without copper monitoring.

Lipid Peroxidation, Plain-English Note

Lipid peroxidation is a free-radical chain reaction where reactive molecules steal electrons from lipids in cell membranes. That can damage membrane structure, change membrane fluidity, generate toxic byproducts, and contribute to inflammatory and ferroptosis-related pathways. The support-stack goal is not to pretend oxidation can be eliminated. The practical goal is to reduce unnecessary oxidative burden while labs and symptoms are monitored.

Alcohol Dehydrogenase Note

Alcohol dehydrogenase is an enzyme group found primarily in the liver and stomach lining. It catalyzes the conversion of ethanol into acetaldehyde, the first major step in alcohol metabolism. Zinc is one relevant cofactor, but alcohol avoidance or strict minimization is the cleaner move during a complex metabolic protocol.

Blood Work and Imaging Before Starting

Baseline testing gives the protocol a scoreboard. The goal is to detect metabolic dysfunction, hepatic stress, biliary risk, nutrient issues, and cardiovascular risk before adding signal-heavy compounds.

Baseline item	Baseline item
Fasting insulin	Fasting glucose
HOMA-IR: fasting glucose in mg/dL x fasting insulin in uIU/mL / 405	ALT
AST	ALP: alkaline phosphatase
GGT	Total bilirubin
Direct bilirubin	Albumin
Total protein	Triglycerides
HDL	LDL
Ferritin	Complete metabolic panel
Lipid panel	HbA1c
Vitamin D	B12
Folate	Selenium
Liver ultrasound with elastography for baseline hepatic steatosis and fibrosis	

Repeat Labs Every 5-6 Weeks During Retatrutide Escalation

Repeat lab	Why it matters
Fasting insulin	Metabolic response and insulin-resistance trend.
Fasting glucose	Useful with fasting insulin for HOMA-IR trend.
GGT	Sensitive hepatic/biliary stress marker. Interpret with provider and other liver markers.
ALT and AST	Hepatocellular stress trend.
Triglycerides	Lipid export/metabolic trend.
HDL	Needed for triglyceride/HDL ratio.
Triglyceride/HDL ratio	Practical trend marker for insulin resistance and lipid handling.

Provider-Review Adjustment Triggers

Safety framing These are not self-directed orders. They are pre-defined provider-review triggers. Any dose reduction, hold, medication change, or

additional testing should be cleared with the clinician supervising the protocol.	
Trigger	Action for provider discussion
GGT rising or not improving	Treat as a hepatic/biliary stress signal. Discuss holding or reducing retatrutide, commonly by about 50% in a conservative plan, and reassess labs in 5-6 weeks.
Triglyceride/HDL ratio rising	Treat as possible worsening lipid handling or insulin-resistance signal. Discuss protein intake, L-carnitine adequacy, retatrutide dose reduction/hold, and reassessment in 5-6 weeks.
Persistent nausea, vomiting, reflux, constipation, diarrhea, dehydration, abdominal pain, or right-upper-quadrant pain	Do not escalate retatrutide. Discuss evaluation for pancreatitis, gallbladder/biliary issues, dehydration, electrolyte imbalance, or dose intolerance.
Resting heart rate rising meaningfully or palpitations	Pause escalation and review. Retatrutide trials reported dose-dependent heart-rate increases.
Sleep, mood, anxiety, or cognitive changes	Do not add cognitive peptides. Stabilize the current stack first.

Daily Tracking Template

Date	Compounds and doses	Energy AM/PM	Sleep hours + quality	Mood/clarity	GI/appetite	Weight	Notes/sides

Weekly Review Template

Week	New variable added?	Best signal	Worst signal	Labs/biometrics	Decision: continue, hold, reduce, or provider review
1					
2					
3					
4					
5					
6					
7					
8					

Evidence and Safety Notes

- Retatrutide: investigational triple agonist. Phase 2 obesity data reported dose-related gastrointestinal adverse events and dose-dependent heart-rate increases. Treat it as the most signal-dominant compound in the plan.
- BPC-157, MOTS-c, Semax, Selank, and related compounded peptides: FDA has identified safety-risk concerns for certain bulk substances, including immunogenicity, impurity, API characterization, and limited human exposure information for some compounds.
- Vitamin E: NIH ODS describes antioxidant roles, but also notes bleeding-related cautions, medication interactions, and mixed trial outcomes at supplemental doses around 400 IU/day.
- Selenium: NIH ODS lists the adult tolerable upper intake level as 400 mcg/day. A 200 mcg supplement can be reasonable to discuss, but total intake matters.
- Ursodiol/TUDCA family: ursodeoxycholic acid is used for cholesterol gallstones and cholestatic liver diseases. TUDCA should be framed as a bile-acid support candidate, not guaranteed protection against biliary sludge or cholestasis.

Selected References

- NIH Office of Dietary Supplements. Vitamin E Fact Sheet for Health Professionals. <https://ods.od.nih.gov/factsheets/VitaminE-HealthProfessional/>
- NIH Office of Dietary Supplements. Selenium Fact Sheet for Health Professionals. <https://ods.od.nih.gov/factsheets/Selenium-HealthProfessional/>
- NCBI LiverTox. Ursodiol, Ursodeoxycholic Acid. <https://www.ncbi.nlm.nih.gov/books/NBK548309/>
- FDA. Certain Bulk Drug Substances for Use in Compounding that May Present Significant Safety Risks. <https://www.fda.gov/drugs/human-drug-compounding/certain-bulk-drug-substances-use-compounding-may-present-significant-safety-risks>
- Jastreboff AM, et al. Triple-Hormone-Receptor Agonist Retatrutide for Obesity. New England Journal of Medicine, 2023. <https://www.nejm.org/doi/full/10.1056/NEJMoa2301972>